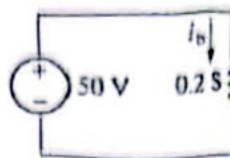
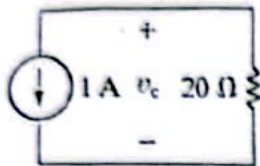




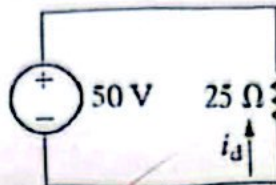
(a)



(b)



(c)



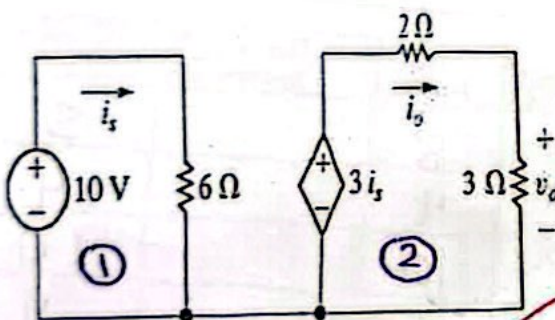
(d)

Q1: Calculate the following values for the above circuits: (8 marks)

- (a) $v_a = \dots 8V \dots$, $p_{8\Omega} = \dots 8W \dots$
 (b) $i_b = \dots 1A \dots$, $p_{0.2S} = \dots 500W \dots$
 (c) $v_c = \dots 20V \dots$, $p_{20\Omega} = \dots 20W \dots$
 (d) $i_d = \dots 2A \dots$, $p_{25\Omega} = \dots 100W \dots$

+8

Q2:- Use KVL & Ohm's law to find the voltage v_o . (4 marks)



$$\text{loop ①} \quad -10 + 6i_s = 0$$

$$6i_s = 10 \rightarrow i_s = 1.67A$$

$$\text{loop ②} \quad -3i_s + 2i_o + 3i_o = 0$$

$$-3(1.67) + 8i_o = 0$$

$$8i_o = 5$$

$$i_o = 1A$$

$$v_o = i_o(3)$$

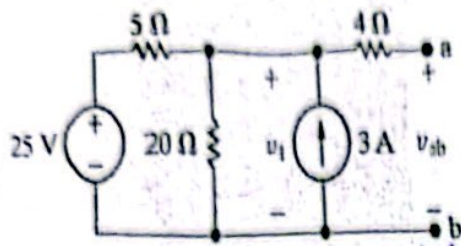
$$= (1)(3) = 3V$$

Solution:

+4

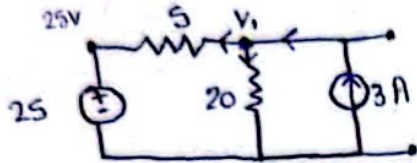


Q3: Find the Thevenin equivalent voltage of the circuit in Fig and calculate R_{TH} using: V_{TH}/I_{SC} (10 marks)



nodal analysis

1. Calculating V_{TH} :

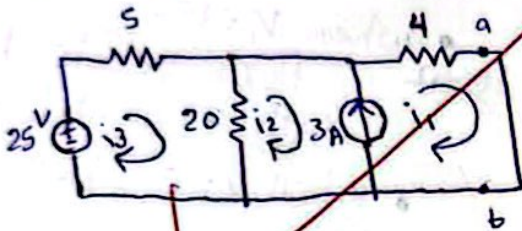


$$3 = \frac{1}{20}V_1 + \frac{1}{5}V_1 - \frac{25}{5}$$

$$8 = \frac{1}{4}V_1$$

$$V_1 = 32V = V_{ab} = V_{TH}$$

2. Finding R_{TH} , using $R_{TH} = V_{TH} / I_{SC}$



$$i_1 = i_{SC}$$

$$i_1 - i_2 = 3 \quad \text{--- (1)}$$

$$4i_1 + 20i_2 - 20i_3 = 0 \quad \text{--- (2)}$$

$$5i_3 + 20i_3 - 20i_2 = 25 \quad \text{--- (3)}$$

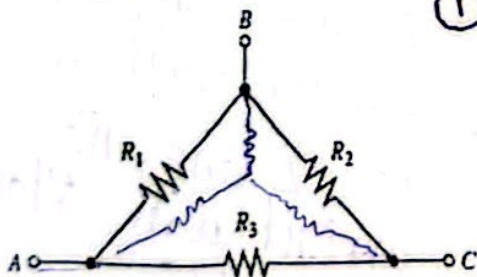
$$i_1 = 4A \quad i_2 = 1A \quad i_3 = 1.8A$$

$$R_{TH} = \frac{V_{TH}}{I_{SC}} = \frac{32}{4} = 8\Omega$$

Q4: Describe the total resistance between each pair of terminals in Figure

1. From A to C:

2. From B to C:



$$\textcircled{1} R_{eq} = \frac{(R_1 + R_2) R_3}{R_3 + (R_1 + R_2)} \rightarrow R_1, R_2 \text{ in series, Parallel with } R_3$$

$$\textcircled{2} R_{eq} = \frac{(R_1 + R_3) R_2}{(R_1 + R_3) + R_2}$$

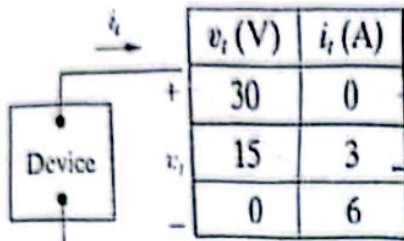
R_1, R_3 in series, in Parallel with R_2

(4 marks)



Q5:

(4 marks)



(a)

(b)

~~$V = 30$~~

$$V = 3i$$

$$R = \frac{15}{3} = 5 \Omega$$

$$15 = 3 \times i$$

$$i = 5$$

$$\frac{30 - 15}{0 - 3} = -5 \Omega$$

$$2V_1 = -\frac{1}{2} i_1$$

$$V_1 = -\frac{1}{4} i_1$$

- Construct a circuit model of the device inside the box.
- Using this circuit model, predict the power this device will deliver to a 10Ω resistor.

1. The circuit:

- when $V_1 = 30V$

There's an open circuit

- when $V_1 = 15V$

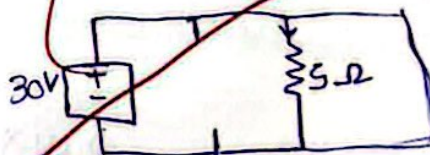
There's a resistor $R = 5 \Omega$

with a current $i = 3A$

- when $V = 0$, $i = 6$

Short circuit

2. $p =$



even if we replace the resistor with a 10Ω one, no current will flow through it, since the short circuit

$P = 0W$

GOOD LUCK